

BCNF Check Table

Relation	FD	Left SPK?	BCNF?
User	MAV_ID $\rightarrow$...	Yes (PK)	Yes
User	Username $\rightarrow$...	Yes (CK)	Yes, so username gets a UNIQUE constraint
User	Email $\rightarrow$...	Yes (CK)	Yes, so email gets a UNIQUE constraint
Homecook	HOMECOOK_MAV_ID $\rightarrow$ {}	Yes (PK)	Yes
Chef	CHEF_MAV_ID $\rightarrow$ chef_id	Yes (PK)	Yes
Chef	chef_id $\rightarrow$ CHEF_MAV_ID	Yes (CK)	Yes, so chef_id gets a UNIQUE constraint
Cookbook	COOKBOOK_ID $\rightarrow$ cookbook_name, date_of_creation, description, primary_cuisine, chef_mav_id	Yes (PK)	Yes
Merchant	REGISTRATION_NUMBER $\rightarrow$ merchant_name, website, number_of_employees, headquarter_city	Yes (PK)	Yes
Merchant	Website $\rightarrow$ REGISTRATION_NUMBER, merchant_name, number_of_employees, headquarter_city	Yes (CK)	Yes, so website gets a UNIQUE constraint
Recipe	{COOKBOOK_ID, RECIPE_NUMBER} $\rightarrow$ title, source_url, date_of_publish, time_of_publish...	Yes (PK)	Yes
Campaign	{MERCHANT_REGISTRATION_NUMBER, CAMPAIGN_NUMBER} $\rightarrow$ advertised_grocery_item_name...	Yes (PK)	Yes
Subscribe	{HOMECOOK_MAV_ID, COOKBOOK_ID} $\rightarrow$ {}	Yes (PK)	Yes
Follow/Unfollow	{HOMECOOK_MAV_ID, CHEF_MAV_ID} $\rightarrow$ {}	Yes (PK)	Yes
Like/Dislike	{HOMECOOK_MAV_ID, COOKBOOK_ID, RECIPE_NUMBER} $\rightarrow$ interaction_type	Yes (PK)	Yes
Read	{HOMECOOK_MAV_ID, COOKBOOK_ID, RECIPE_NUMBER, start_time} $\rightarrow$ {end_time, location}	Yes (PK)	Yes
Display_On	{COOKBOOK_ID, RECIPE_NUMBER, MERCHANT_REGISTRATION_NUMBER, CAMPAIGN_NUMBER} $\rightarrow$ {display_cost}	Yes (PK)	Yes

Dependency levels:

L0:

- User (No FK)
- Merchant (No FK)

L1:

- Homecook ($\rightarrow$ User)
- Chef ($\rightarrow$ User)
- Campaign ($\rightarrow$ Merchant)

L2:

- Cookbook ($\rightarrow$ Chef)
- Follow/Unfollow ($\rightarrow$ Homecook, $\rightarrow$ Chef)

L3:

- Subscribe ($\rightarrow$ Homecook, $\rightarrow$ Cookbook),
- Recipe ($\rightarrow$ Cookbook)

L4:

- Like/Dislike ($\rightarrow$ Homecook, $\rightarrow$ Recipe via composite FK (COOKBOOK_ID, RECIPE_NUMBER))
- Read ($\rightarrow$ Homecook, $\rightarrow$ Recipe via composite FK (COOKBOOK_ID, RECIPE_NUMBER))
- Display_On $\rightarrow$ Recipe via composite FK (COOKBOOK_ID, RECIPE_NUMBER), $\rightarrow$ Campaign via composite FK (MERCHANT_REGISTRATION_NUMBER, CAMPAIGN_NUMBER))

Execution Order:

- Creation Order (projectDBcreate.sql): $L0 \rightarrow L1 \rightarrow L2 \rightarrow L3 \rightarrow L4$
- Drop Order (projectDBdrop.sql): $L4 \rightarrow L3 \rightarrow L2 \rightarrow L1 \rightarrow L0$